

The Core of Energy Balance

- Balance between calories consumed and calories expended.
- Positive balance means surplus energy is stored.
- Negative balance means drawing from stored energy.
- Energy balance is NOT determined solely by exercise.



Energy In: Calories consumed through food and drink.



Energy Out: Calories burned through resting biological functions and physical activity.

Basal Metabolic Rate (BMR)

- Absolute minimum energy required at complete rest.
- Powers vital organs, breathing, and circulation.
- Analogy: A smartphone draining battery on standby mode.
- Does NOT include energy for digestion or physical movement.



Like a phone on standby, BMR is the background energy required to keep your body's 'operating system' running.

What Influences BMR?

- Age: Metabolism naturally slows as cellular processes age.
- Sex: Biological factors alter baseline hormonal needs.
- Composition: Muscle requires more resting energy than fat.
- BMR is highly individualized, not a fixed universal number.



Age: BMR naturally decreases as cellular processes slow down over time.



Sex: Biological and hormonal differences affect baseline caloric requirements.



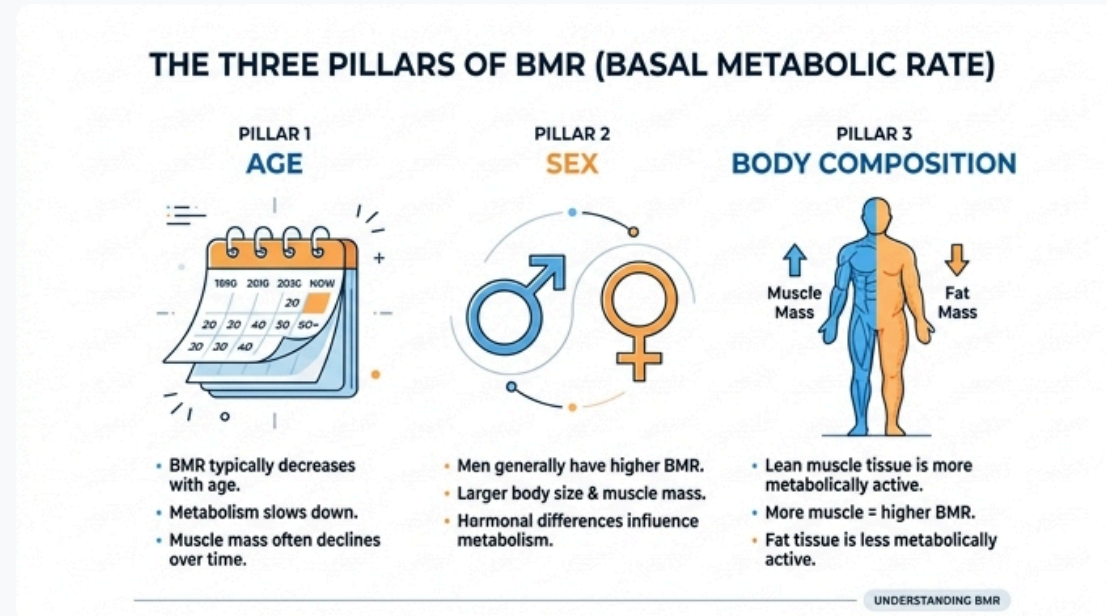
Body Composition: Muscle tissue is highly metabolically active compared to fat.

Calculating Your BMR

- Standard formulas provide a reliable baseline estimation.
- The Mifflin-St Jeor equation is the current clinical standard.
- For Men: $BMR = 10m + 6.25h - 5a + 5$
- For Women: $BMR = 10m + 6.25h - 5a - 161$

Factors Influencing BMR

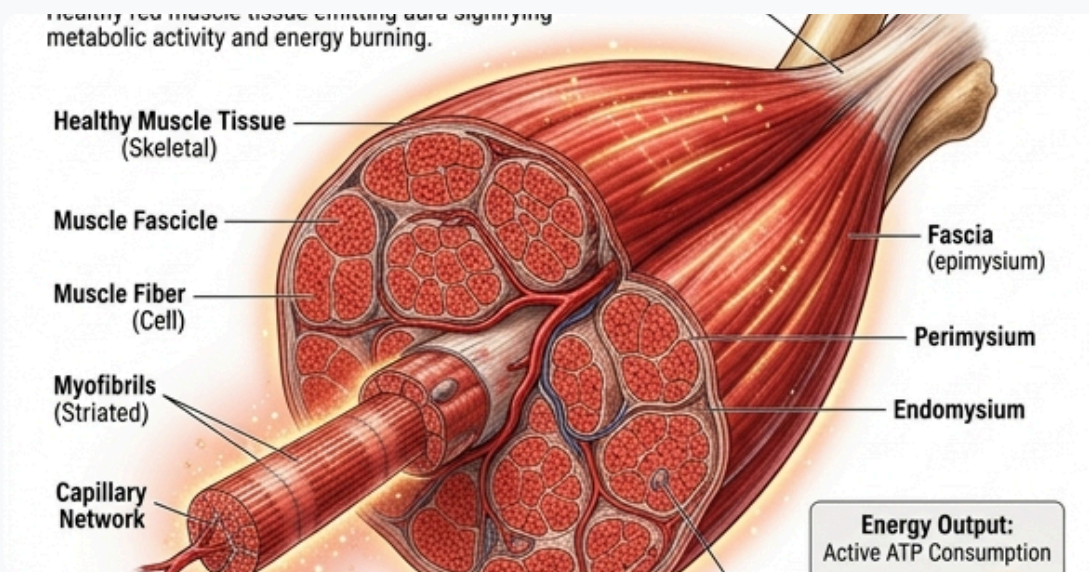
- BMR is highly individualized
- Modeled conceptually as $BMR = f(Age, Sex, Composition)$
- Baseline energy needs are dynamic, not static



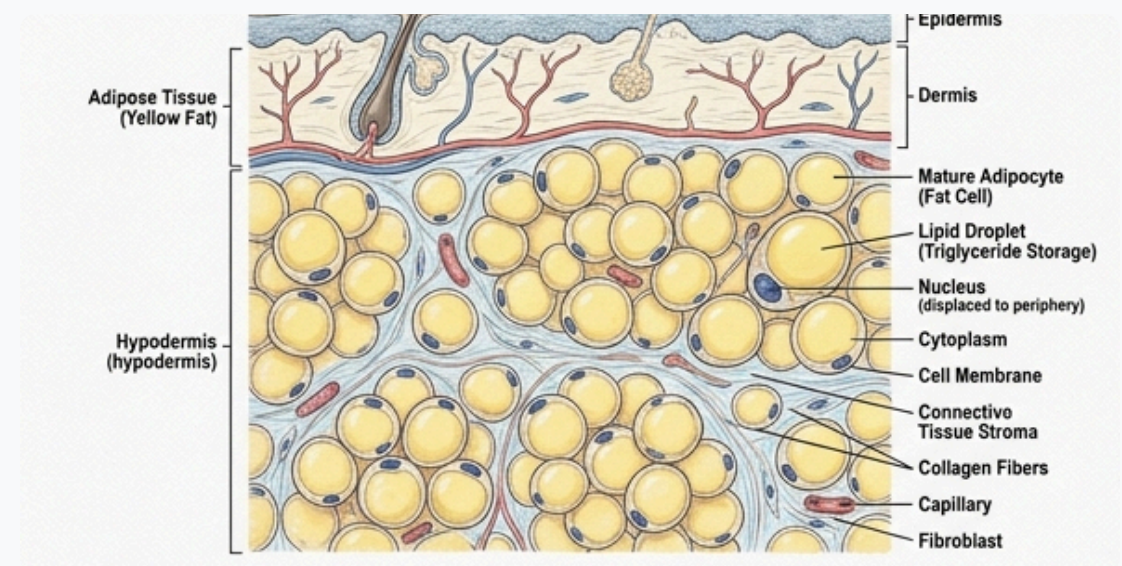
Age, sex, and body composition are the primary drivers that determine your Basal Metabolic Rate.

Body Composition: Muscle vs. Fat

- Muscle is a metabolically active tissue
- Fat requires significantly less baseline energy
- Relationship: $MuscleMass \propto BMR$



Muscle tissue actively burns calories even when the body is completely at rest.



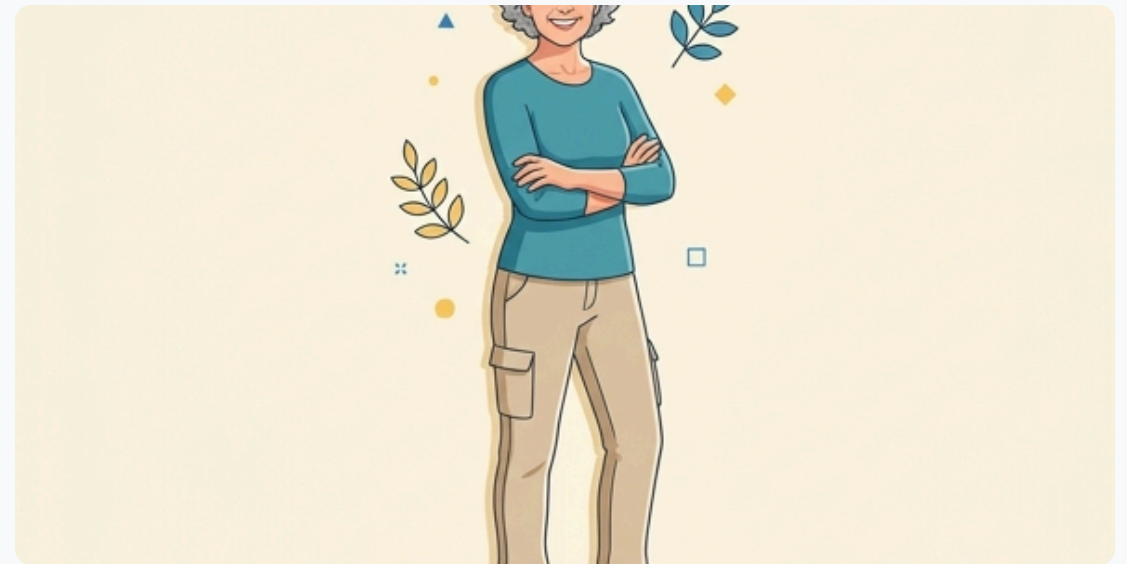
Adipose tissue acts primarily as storage and requires very little energy to maintain.

Comparing BMR Profiles

- Age naturally decreases metabolic rate
- Biological sex impacts baseline muscle density
- Example: 25-year-old male vs. 65-year-old female



Youth and naturally higher muscle density result in a significantly higher BMR.



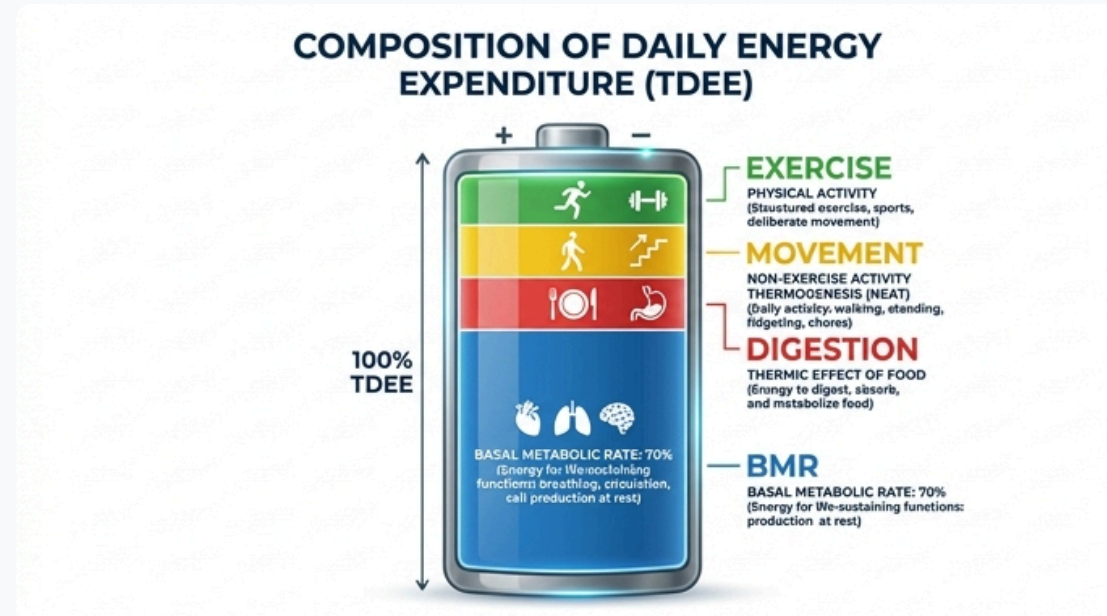
Older age and naturally lower muscle mass result in a lower baseline energy requirement.

Misconception: BMR is Fixed

- Myth: You are born with an unchangeable metabolism
- Truth: BMR adapts to your physical state
- Resistance training increases BMR over time
- BMR is a dynamic biological baseline

What BMR is NOT

- BMR is strictly your body's baseline survival energy.
- It excludes any physical movement or physical activity.
- It excludes the energy used to digest food.



BMR only makes up the foundational portion of your total daily energy expenditure.

Three Things Excluded from BMR

- Digesting and processing food.
- Daily tasks like walking or doing chores.
- Dedicated workouts and sports.



Digesting food requires energy, but this is NOT part of BMR.



Incidental daily movements are active calories, NOT BMR.



Structured exercise is an active caloric burn added on top of BMR.

Baseline vs. Active Energy

- BMR happens even if you sleep all day.
- Active energy requires muscular effort.
- Example: Burning 400 calories jogging is active energy.
- Active calories are added on top of BMR.



BMR: Energy burned while completely at rest.



Active Energy: Additional calories burned during physical exertion.

Calculating Total Energy Expenditure

- Total Daily Energy Expenditure (TDEE) is the complete picture.
- Formula: $TDEE = BMR + ActiveEnergy + Digestion$
- Do not conflate the baseline with the total.

Standard Formulas for Estimating *BMR*

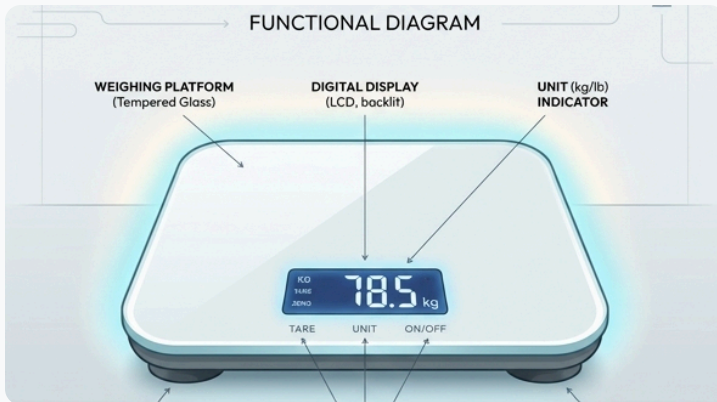
- Nutritionists use standard equations to estimate baseline energy.
- The Mifflin-St Jeor equation is the primary industry standard.
- Calculations provide a starting point for personalized nutrition plans.



Nutritionists use standardized equations to estimate metabolic baselines without needing laboratory equipment.

Variables of the Mifflin-St Jeor Equation

- Weight: Total body mass drives overall energy demand.
- Height: Determines total body surface area.
- Age and Sex: Accounts for biological shifts in muscle mass.



Weight indicates total mass that requires energy to maintain.



Height helps determine the body's overall surface area.



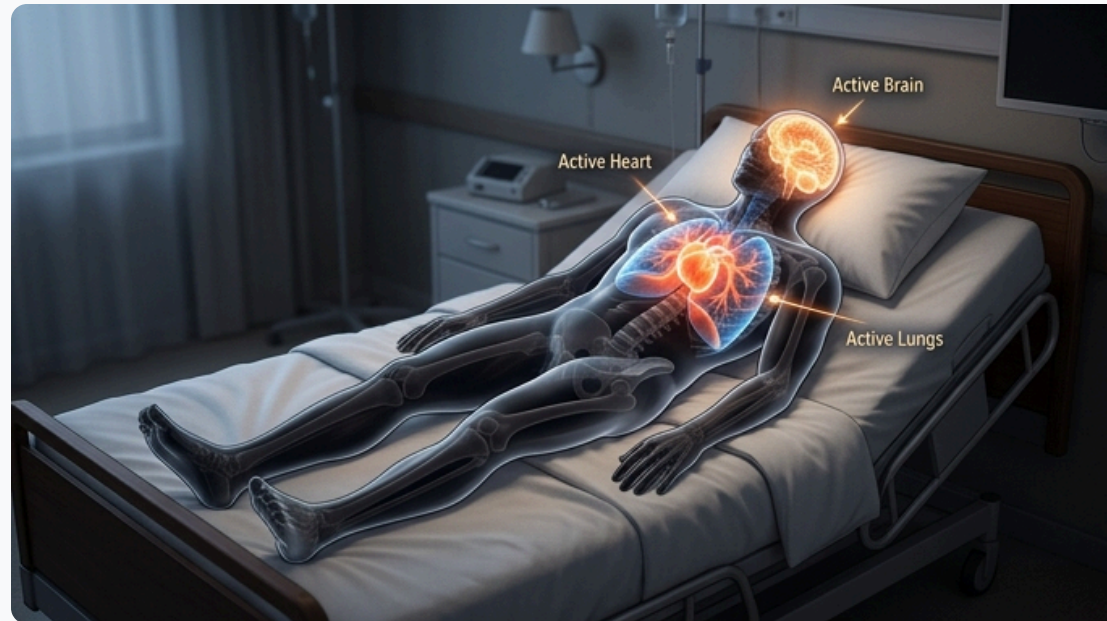
Age and sex account for natural metabolic and hormonal shifts.

A Real-World Calculation

- Subject Profile: 30-year-old male, 70 kg weight, 170 cm height.
- Applying the variables to the Mifflin-St Jeor equation yields total resting energy.
- Calculated baseline: $BMR \approx 1650$ kcal/day.

Visualizing the Baseline

- Represents energy solely for vital internal organ function.
- Equivalent to 24 hours of complete, motionless bed rest.
- Non-example: Does NOT include energy for walking, digesting, or talking.



BMR only accounts for the energy required to keep your vital organs functioning at complete rest.

Scenario: Identical Stats, Different Lifestyles

- Same age, height, and overall weight.
- Woman A: Sedentary office worker.
- Woman B: Active weightlifter.
- Question: Who burns more calories at rest?



Woman A: Sedentary lifestyle with lower muscle mass.



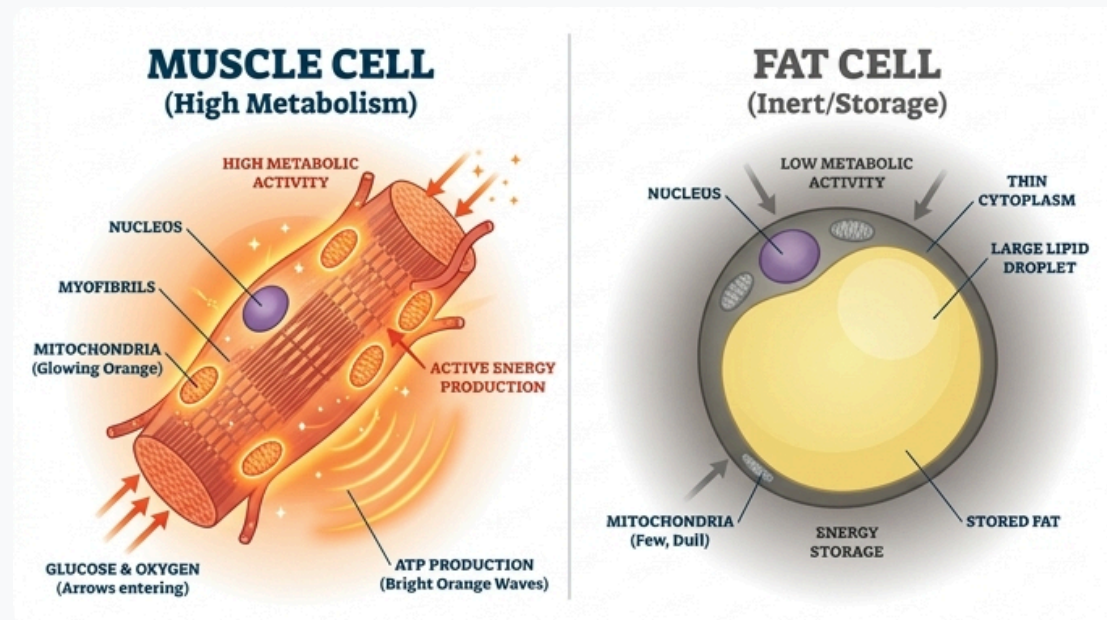
Woman B: Active lifestyle with higher muscle mass.

The Role of Body Composition

- Weight alone does not dictate BMR .
- Muscle tissue is highly metabolically active.
- Fat tissue requires significantly less energy.
- Body composition shifts the energy equation.

Muscle Drives Baseline Energy

- Woman B has a definitively higher *BMR*.
- More muscle means higher daily baseline energy needs.
- More calories are burned automatically while resting.
- Note: This is baseline energy, not active workout burn.



Muscle tissue constantly demands baseline energy to survive, directly increasing BMR.